

ServiceNow Incident Management Project Documentation

1. Introduction

Project Title:

ServiceNow Incident Management Workflow Automation

Platform Used:

ServiceNow Service Operations Workspace (Australia Version)

Project Description:

This project focuses on implementing an end-to-end Incident Management workflow in ServiceNow. The project covers incident creation, assignment, escalation, child incident handling, Change Request integration, Knowledge Management, and SLA monitoring.

2. Project Overview

Purpose:

The purpose of this project is to manage IT service incidents efficiently using ServiceNow ITSM capabilities. It demonstrates the complete incident lifecycle from issue reporting to final resolution.

Features:

- Incident creation and management
- Incident assignment to support teams
- Incident escalation process
- Child incident management
- Configuration Item (CI) tracking
- Change Request integration
- Knowledge article creation
- SLA monitoring
- Incident resolution workflow
- Multi-team collaboration

3. Architecture

ServiceNow Platform:

The project is developed on the ServiceNow cloud platform using IT Service Management (ITSM) modules.

Incident Management:

Handles the complete lifecycle of incidents including creation, investigation, assignment, updates, and resolution.

Change Management:

Used to create and manage Change Requests related to incidents requiring system modifications.

Knowledge Management:

Used to create and publish knowledge articles from resolved incidents for future reference.

Configuration Management Database (CMDB):

Stores and manages Configuration Items such as services, applications, and infrastructure components.

Service Operations Workspace:

Provides a centralized interface for agents to manage incidents, tasks, SLAs, and related records.

4. Setup Instructions

Prerequisites:

- ServiceNow instance access
- Service Operations Workspace access
- ITSM roles and permissions
- Required users and groups configured
- Required services and Configuration Items created

Environment Setup:

1. Login to the ServiceNow instance.
2. Open Service Operations Workspace.
3. Configure required services and service offerings.

4. Create required Configuration Items in CMDB.
5. Create users and assign appropriate roles.
6. Start Incident Management workflow execution.

Configuration Setup:

Created Service:

- Remote Access

Created Service Offering:

- Corporate VPN

Created Configuration Item:

- VPN Gateway / VPN-SRV-02

5. Folder Structure

ServiceNow does not follow a traditional folder structure like application-based projects.

The project is organized using ServiceNow modules:

Incident Management Module:

Contains:

- Incidents
- Child Incidents
- Incident Tasks
- Related Records

Change Management Module:

Contains:

- Change Requests
- Change Activities
- Approvals

Knowledge Management Module:

Contains:

- Knowledge Articles

- Article Categories
- Publishing Workflow

CMDB Module:

Contains:

- Configuration Items
- Services
- Service Offerings

6. Running the Application

The project runs on the ServiceNow cloud environment.

Steps to access the project:

1. Login to ServiceNow instance.
2. Navigate to Service Operations Workspace.
3. Open Incident Management.
4. Create or select an incident.
5. Perform assignment, escalation, change integration, and resolution activities.

7. API Documentation

ServiceNow provides REST APIs for integrating external applications.

Common ServiceNow APIs:

Incident API:

Used for creating and managing incidents.

Operations:

- Create Incident
- Update Incident
- Retrieve Incident Details

Change Request API:

Used for managing change records.

Operations:

- Create Change Request
- Update Change Request Status
- Retrieve Change Details

Knowledge API:

Used for accessing knowledge articles.

Operations:

- Create Article
- Search Articles
- Retrieve Article Details

8. Authentication

ServiceNow provides secure authentication and authorization mechanisms.

Authentication Methods:

- Username and Password authentication
- Role-based access control
- User impersonation for testing different user activities

Authorization:

Access to modules and records is controlled using ServiceNow roles and permissions.

Example:

- Incident users can create and update incidents.
- Network users can handle assigned incidents.
- Knowledge users can create and publish articles.

9. User Interface

The project uses ServiceNow Service Operations Workspace.

Major UI Components:

Incident Workspace:

Used for:

- Creating incidents

- Updating incident details
- Assigning incidents
- Resolving incidents

Related Records:

Used for:

- Child incidents
- Change Requests
- Knowledge articles

Task SLA Section:

Used for monitoring SLA progress.

Agent Assist:

Used for finding and attaching relevant knowledge articles.

Screenshots included:

- Incident creation page
- Incident details page
- Assignment update
- Child incident creation
- Change Request creation
- Knowledge article publishing
- SLA tracking
- Final resolved incident

10. Testing**Testing Strategy:**

The project was tested by validating each stage of the Incident Management lifecycle.

Test Cases:**Incident Creation Testing:**

Verified incident creation with correct user, description, category, and service details.

Assignment Testing:

Verified incident assignment to appropriate groups and users.

Child Incident Testing:

Verified creation and linking of child incidents.

Change Integration Testing:

Verified Change Request creation and linking with incidents.

Knowledge Testing:

Verified knowledge article creation, publishing, and linking.

Resolution Testing:

Verified incident resolution and activity updates.

SLA Testing:

Verified SLA tracking visibility.

11. Screenshots or Demo

Screenshots demonstrate the complete workflow:

1. Service and Configuration Item Setup
2. Incident Creation
3. Assignment Group Update
4. Knowledge Article Attachment
5. Child Incident Creation
6. Configuration Item Update
7. Cause Addition
8. Change Request Creation
9. Incident Resolution
10. Knowledge Article Publishing
11. SLA Tracking
12. Final Validation

12. Known Issues

- ServiceNow UI differs between different versions.
- Some options may not appear depending on user roles.
- Zurich documentation steps may differ from Australia version UI.
- Change Request options may require additional permissions.
- Knowledge publishing workflow may vary based on configuration.

13. Future Enhancements

- Automate incident assignment using workflows.
- Implement AI-based incident classification.
- Add automated notifications and escalations.
- Improve SLA reporting dashboards.
- Create automated problem management workflows.
- Enhance knowledge recommendations using AI.

Conclusion

The ServiceNow Incident Management project successfully demonstrates an end-to-end ITSM workflow. It covers incident handling, escalation, child incident management, Change Request integration, Knowledge Management, and SLA tracking. The implementation improves service delivery, collaboration, and issue resolution efficiency.